Niven Rational and Irrational Benchmark

<table>
<tr><th>Natural language statement</th><th>Domain</th><th>GroundTruth</th></tr>
<tr><td>Any rational fraction a/b is expressible as a terminating decimal or an infinite periodic decimal; conversely, any decimal expansion which is either terminating or infinite periodic is equal to some rational number. (2.4, with 2.5 "Terminating Decimals Written as Periodic Decimals".) Notation: the decimal expansion of x in [0,1) is the digit sequence d_k = floor(10^{k+1} x) mod 10; a terminating expansion is the eventually periodic one whose repeating digit is 0.</td><td>Number theory</td><td>noncomputable def decimalDigit (x : R) (k : N) : N :=
  ((10 : R) ^ (k + 1) * x).toNat % 10

def EventuallyPeriodic (d : N -> N) : Prop :=
  exists N p : N, 0 < p /\ forall k, N <= k -> d (k + p) = d k

theorem niven_2_4_rational_iff_periodic_decimal (x : R) (hx : x in Ico (0 : R) 1) :
    (exists q : Q, x = q) <-> EventuallyPeriodic (decimalDigit x) := by
  sorry</td></tr>
<tr><td>3.5. sqrt(2) + sqrt(3) is irrational.</td><td>Number theory</td><td>theorem niven_3_5_sqrt_two_add_sqrt_three_irrational :
    Irrational (Real.sqrt 2 + Real.sqrt 3) := by
  sorry</td></tr>
<tr><td>5.3, Example 3. Let c and d be two different non-negative integers. Prove that log(2^c 5^d) is irrational. Notation: all logarithms are to base 10, i.e. log y = k means 10^k = y.</td><td>Number theory</td><td>theorem niven_5_3_log_two_pow_five_pow_irrational (c d : N) (hcd : c != d) :
    Irrational (Real.logb 10 (2 ^ c * 5 ^ d)) := by
  sorry</td></tr>
<tr><td>Theorem on Geometric Constructions. Beginning with a line segment of unit length, any length that can be constructed by straightedge and compass methods is an algebraic number of degree 1, or 2, or 4, or 8, ..., i.e., in general, an algebraic number whose degree is a power of 2. Notation: a length is constructible when it can be obtained from a unit segment by straightedge and compass; algebraically these are the numbers reachable from the rationals by the field operations and by square roots of non-negative constructed quantities. The degree of an algebraic number is the degree of its minimal polynomial over the rationals.</td><td>Number theory</td><td>inductive IsConstructible : R -> Prop
  | rat (q : Q) : IsConstructible q
  | add {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x + y)
  | neg {x : R} : IsConstructible x -> IsConstructible (-x)
  | mul {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x * y)
  | inv {x : R} : IsConstructible x -> IsConstructible x^-1
  | sqrt {x : R} : IsConstructible x -> 0 <= x -> IsConstructible (Real.sqrt x)

theorem niven_5_5_constructible_degree_is_power_of_two (x : R) (hx : IsConstructible x) :
    IsAlgebraic Q x /\ exists k : N, Module.finrank Q (Algebra.adjoin Q ({x} : Set R)) = 2 ^ k := by
  sorry</td></tr>
<tr><td>5.5. The duplication of the cube amounts to constructing a line of length 2^{1/3} from a given unit length. Since 2^{1/3} is an algebraic number of degree 3, by the Theorem on Geometric Constructions it is not constructible. Hence it is impossible to duplicate the cube.</td><td>Number theory</td><td>inductive IsConstructible : R -> Prop
  | rat (q : Q) : IsConstructible q
  | add {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x + y)
  | neg {x : R} : IsConstructible x -> IsConstructible (-x)
  | mul {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x * y)
  | inv {x : R} : IsConstructible x -> IsConstructible x^-1
  | sqrt {x : R} : IsConstructible x -> 0 <= x -> IsConstructible (Real.sqrt x)

theorem niven_5_5_duplication_of_the_cube_impossible :
    not IsConstructible ((2 : R) ^ ((1 : R) / 3)) := by
  sorry</td></tr>
<tr><td>5.5. Squaring the circle amounts to constructing a square of area equal to that of a circle of unit radius, i.e. to constructing the length sqrt(pi). Granted that pi is a transcendental number, sqrt(pi) is not an algebraic number of degree a power of 2, so by the Theorem on Geometric Constructions it is not constructible.</td><td>Number theory</td><td>inductive IsConstructible : R -> Prop
  | rat (q : Q) : IsConstructible q
  | add {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x + y)
  | neg {x : R} : IsConstructible x -> IsConstructible (-x)
  | mul {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x * y)
  | inv {x : R} : IsConstructible x -> IsConstructible x^-1
  | sqrt {x : R} : IsConstructible x -> 0 <= x -> IsConstructible (Real.sqrt x)

theorem niven_5_5_squaring_the_circle_impossible (hpi : Transcendental Q Real.pi) :
    not IsConstructible (Real.sqrt Real.pi) := by
  sorry</td></tr>
<tr><td>5.5. To establish that the trisection of an angle is impossible, it is enough to show that a specific angle cannot</td><td>Number theory</td><td>inductive IsConstructible : R -> Prop
  | rat (q : Q) : IsConstructible q</td></tr>
</table>

Natural language statement	Domain	GroundTruth

be trisected by the prescribed methods. The specific angle that we take is 60 degrees. To trisect an angle of 60 degrees means the construction of a 20 degree angle, equivalently of the length cos 20 degrees, which is not constructible.		`  | add {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x + y)`
		`  | neg {x : R} : IsConstructible x -> IsConstructible (-x)`
		`  | mul {x y : R} : IsConstructible x -> IsConstructible y -> IsConstructible (x * y)`
		`  | inv {x : R} : IsConstructible x -> IsConstructible x^-1`
		`  | sqrt {x : R} : IsConstructible x -> 0 <= x -> IsConstructible (Real.sqrt x)`
		
		`theorem niven_5_5_trisection_of_the_angle_impossible :`
		`    not  IsConstructible (Real.cos (Real.pi / 9)) := by`
		`  sorry`

Theorem 6.2. Corresponding to any irrational number alpha there is a unique integer m such that -1/2 < alpha - m < 1/2.	Number theory	`theorem niven_6_2_unique_nearest_integer (a : R) (ha : Irrational a) :`
		`    exists! m : Z, -(1 / 2 : R) < a - m /\ a - m < 1 / 2 := by`
		`  sorry`

7.5. The number e is transcendental. Notation: a real number is transcendental when it satisfies no polynomial equation with rational (equivalently, integer) coefficients, other than the zero polynomial.	Number theory	`theorem niven_7_5_transcendence_of_e : Transcendental Q (Real.exp 1) := by` `  sorry`

Theorem C.5. The set of real transcendental numbers is uncountable. Notation: a real number is algebraic when it is a root of a non-zero polynomial with rational coefficients, and transcendental otherwise.	Number theory	`theorem niven_C_5_transcendentals_uncountable :`
		`    not  {x : R | Transcendental Q x}.Countable := by`
		`  sorry`